A MODELS, PANELS AND THE TRANSFER MATRIX

Seed pair
- R32 · 3 runs
- V-scr · 3 runs

CIFAR-100: 3 + 3 runs

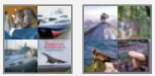
Architecture × init.

scratch / IN init.

V-scr 3	V-IN 2
R224-scr 2	R224-IN 2

runs per cell, CIFAR-10

Reference panels



C10 7 + 1
C100 7 + 2
IN-V2 6 + 1

robust CNNs + secondary

Transfer matrix

20 / 15 / 14 models; target-correct, PGD-10

Exact split

$\log T = \log w + \log \tau + \tfrac{1}{2} r$

w vulnerability, τ symmetric similarity, r direction, per ordered pair

B FROM PAIR SPLIT TO CONTRASTS

panel means
σ symmetric, ρ directional, ω = log w

contrasts
source gap = σ part + ρ part; 2 × 2 cell contrasts

input gradients
cosine c on clean images, Mantel with log τ

σ similarity | ρ direction | ω vulnerability

C REGISTERED BEFORE MEASUREMENT

panel family	K-a, K-b, K-c	CIFAR-10
two-by-two family	H1, H2, H3	CIFAR-10
attack side	best attack per source	
masking check	AutoAttack	
replication	K-a, K-b, K-c	CIFAR-100
gradient family	Mantel tests	all three
ImageNetV2 panel	descriptive, no decision	

CONVENTIONS

- Protocol layer: conditioning sets nested, raw, target-correct, both-correct, successful-source
- h = −log(1 − T) scale; RobustBench checkpoints frozen before measurement